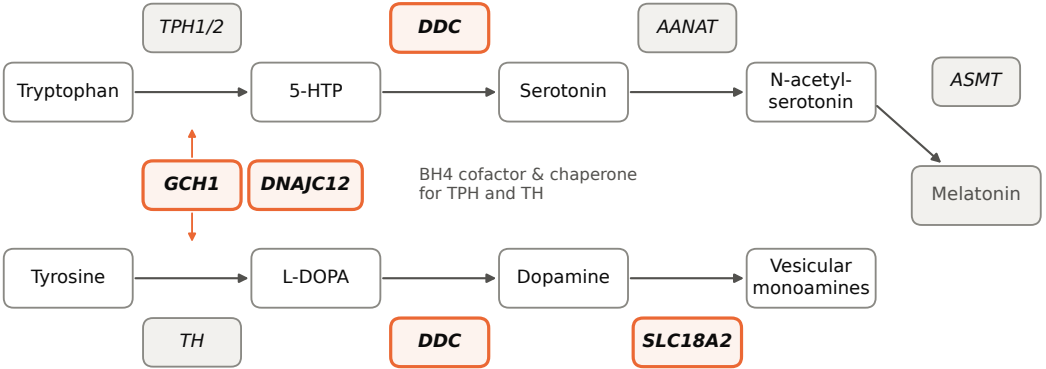

gene inside a chronotype GWAS locus ($r^2 \geq 0.6$ interval); all four are inherited monoamine neurotransmitter disorder genes (GTPCH deficiency / dopa-responsive dystonia, AADC deficiency, DNAJC12 deficiency, VMAT2 deficiency). Grey: context only; no melatonin-pathway gene lies in a locus.

Gene-level support: DDC: top L2G gene (0.95); MAGMA $p < 10^{-15}$; plasma pQTL (CLPP 0.10) · GCH1: nearest gene; top L2G gene (0.65) DNAJC12: top L2G gene (0.41); nearest gene SIRT1 (itself a circadian regulator) · SLC18A2: no gene-level support beyond locus overlap Exome burden: none of the four ($p \geq 0.12$)